

Chapter 2

Graphs, Limits, and Continuity of Functions

2.1 Plotting Graphs

2.1.1 Basics Plot

In this section, we will discuss how to plot graphs using Sage and how to utilize its various plot options. We will discuss in detail several options that will be useful in our study of calculus. The basic syntax for plotting the graph of a function $y = f(x)$ with x ranging in value from a to b is `plot(f,x,a,b)`.

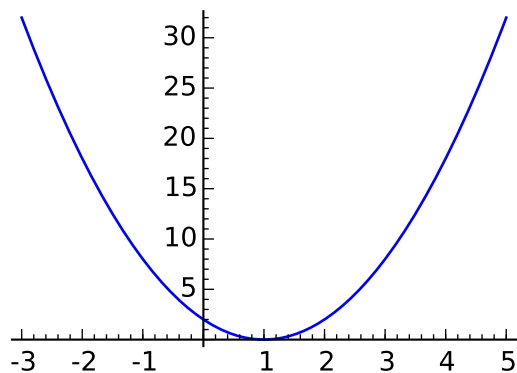
`plot(f(x),x,a,b)`

Example 2.1.1. Plot the graph of $f(x) = 2x^2 - 4x + 2$ along the interval $[-3, 5]$

Solution:

```
sage: g=plot(2*x^2-4*x+2,x,-3,5)
```

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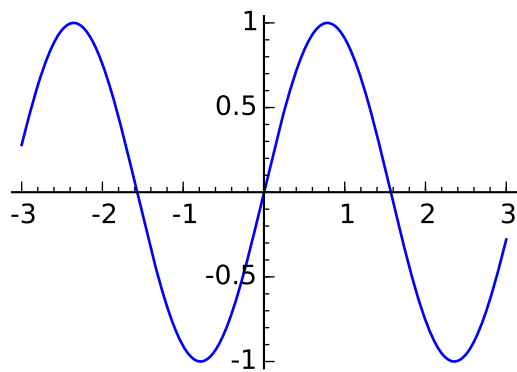


Example 2.1.2. Plot the graph of $y = \sin(2x)$ along the interval $[-3, 3]$

Solution:

```
sage: g=plot(sin(2*x), x, -3, 3)
```

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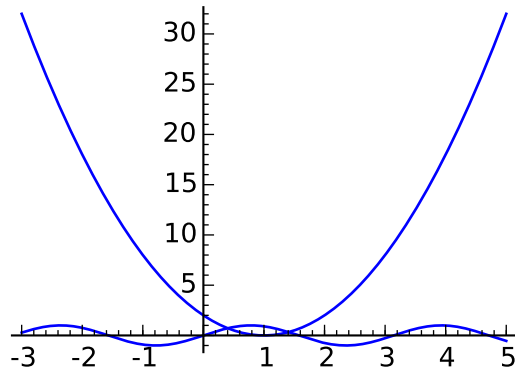


Example 2.1.3. Plot the graphs of the two functions given in Example 1.1 and Example 1.2 prior on the same set of axes to show their points of intersection.

Solution:

```
sage: g=plot((2*x^2-4*x+2, sin(2*x)), x, -3, 5)
```

92

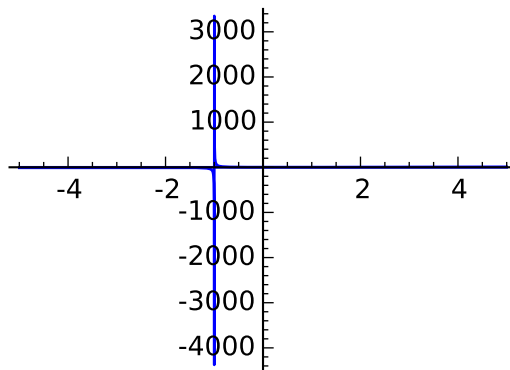


Example 2.1.4. Plot the graphs of $f(x) = \frac{2x^2+x+2}{x+1}$ and $g(x) = \frac{\cos(2x)}{2}$ on the same set of axes.

Solution:

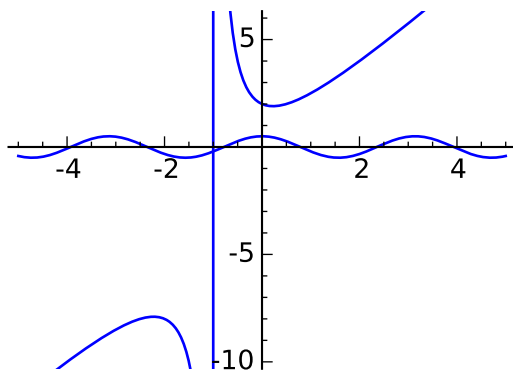
```
sage: g=plot(((2*x^2+x+2)/(x+1)),(cos(2*x))/2),x,-5,5)
```

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Note that the graph of $g(x) = \cos(2x)/2$ is displayed poorly in output above since its range (from -1 to 1) is too small compared to the range of $f(x) = (2x^2 + x + 2)/(x + 1)$. We can zoom in by specify the value of vertical line using **ymin** and **ymax**.

```
sage: g=plot(((2*x^2+x+2)/(x+1)),(cos(2*x))/2),x,-5,5,ymin=-10, 94
        ymax=6)
```



Example 2.1.5. Plot the graphs of the following functions.

(a) $f(x) = \frac{2x^2}{2-x^2}$ (b) $f(x) = 2\sin(x) + \cos(x)$ (c) $f(x) = xe^x + \ln x$ (d) $f(x) = \frac{2x^2}{x^2+2}$

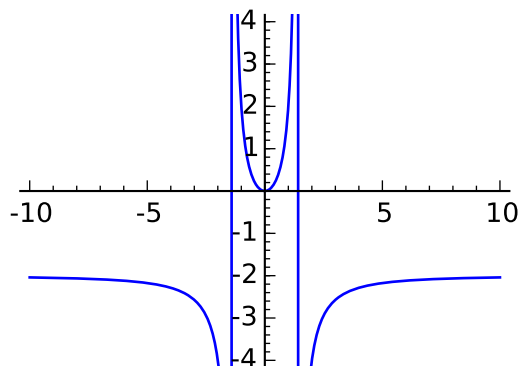
Solution:

We recall that the natural base e is entered as e and that $\ln x$ is **log(x)**. Note that $\sin x$ and $\cos x$ are to be entered as $\sin(x)$ and $\cos(x)$.

(a)

```
sage: g=plot((2*x^2)/(2-x^2),x,-10,10,ymin=-4,ymax=4)
```

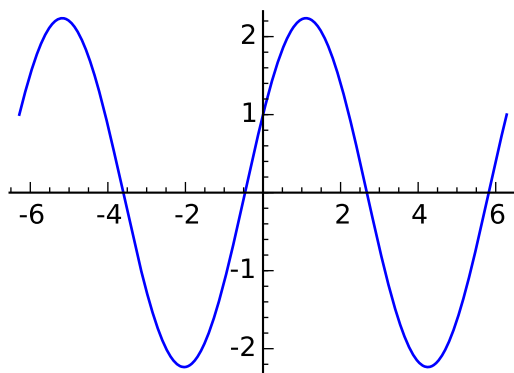
95



(b)

```
sage: g=plot(2*sin(x)+cos(x),x,-2*pi,2*pi)
```

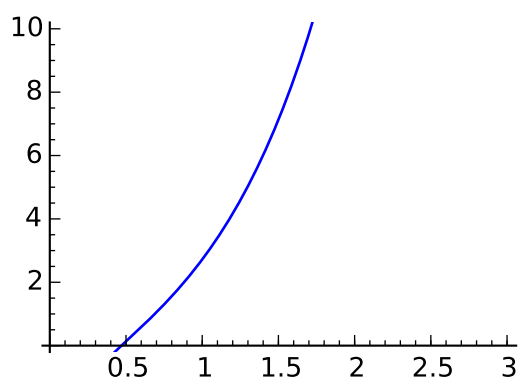
96



(c)

```
sage: g=plot(x*e^x+log(x),x,-3,3,ymin=0,ymax=10)
```

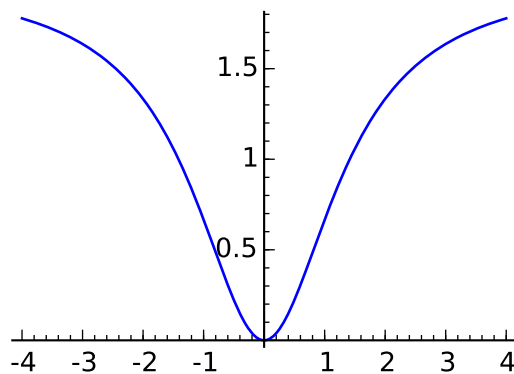
97



(d)

```
sage: g=plot(2*x^2/(x^2+2),x,-4,4)
```

98



2.1.2 Plot Options

Next, we will introduce various options that can be specified within the **plot** command.

- Adding a **title** to a graph:

```
plot(f(x),x,a,b,title="Here is a graph")
```

- Use **figsize** option to control the plot size:

```
plot(f(x),x,a,b,figzie='a number')
```

- Draw a graph with **color**:

```
plot(f(x),x,a,b, color= 'a color')
```

- Draw a graph and specify its **thickness**:

```
plot(f(x),x,a,b,color= 'a color', thickness='a number')
```

- Draw graph with specify the **line style** and **legend_label**:

```
plot(f(x),x,a,b,color= 'a color',linestyle='-', thickness='a number',legend_label='f(x)')
```

- Use **frame** option to puts a box around the graph

```
plot(f(x),x,a,b,frame=True)
```

- Use **axes_labels** to verify the axes:

```
plot(f(x),x,a,b,axes_labels=['x-axis, units','y-axis, units'])
```

- To draw an ellipse, use **implicit_plot** command:

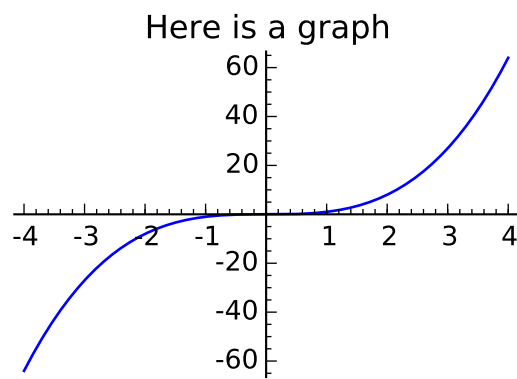
`implicit_plot(f(x),(x,a,b),(y,c,d))`

Example 2.1.6. Plot($x^3, x, -4, 4$) with a title: "Here is a graph"

Solution:

```
sage: g=plot(x^3,x,-4,4, title="Here is a graph")
```

99

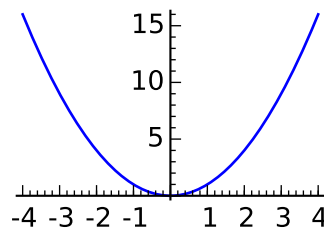


Example 2.1.7. Plot($x^2, x, -4, 4$) with different size.

Solution:

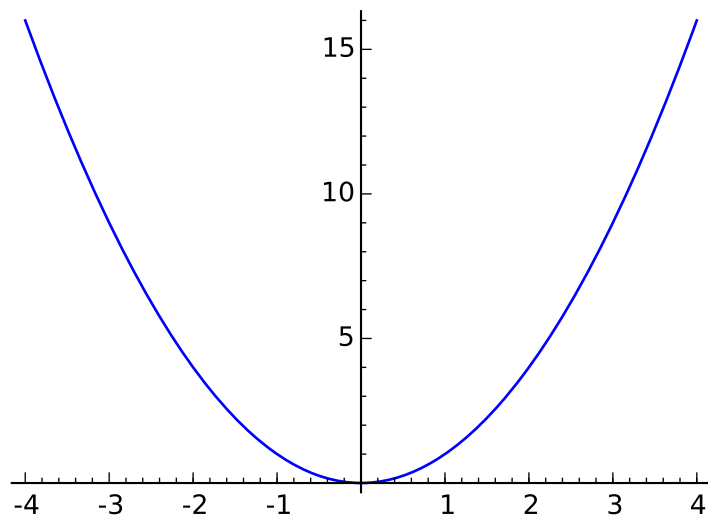
```
sage: g=plot(x^2,x,-4,4,figsize=2)
```

100



```
sage: g=plot(x^2,x,-4,4,figsize=4)
```

101

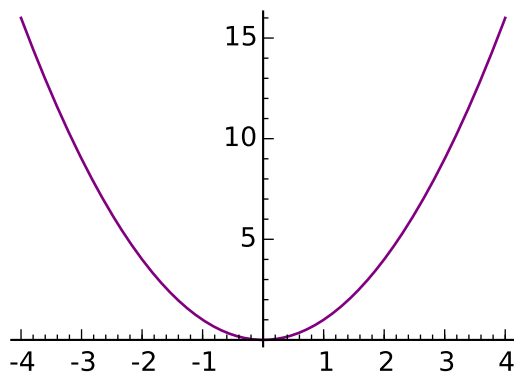


Example 2.1.8. Plot($x^2, x, -4, 4$) with purple color.

Solution:

```
sage: g=plot(x^2,x,-4,4, color= 'purple',figsize=3)
```

102

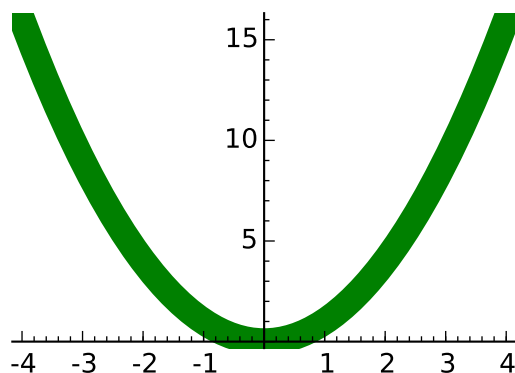


Example 2.1.9. Plot($x^2, x, -4, 4$) with color and thickness features.

Solution:

```
sage: g=plot(x^2,x,-4,4, color= 'green', thickness=10,figsize
=3)
```

103

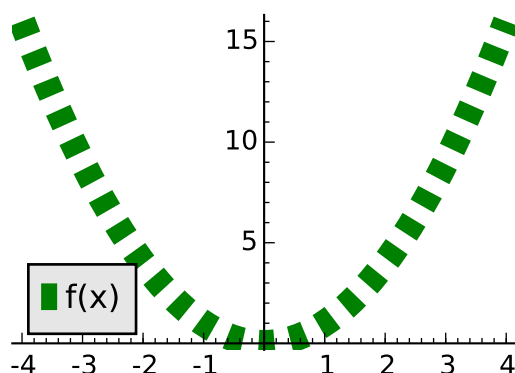


Example 2.1.10. Plot($x^2, x, -4, 4$) with multiple options.

Solution:

```
sage: g=plot(x^2,x,-4,4, color= 'green',linestyle='--',
            thickness=10, legend_label='f(x)',figsize=3)
```

104



Example 2.1.11. Plot multiple function on a single graphic:

Solution:

```
sage: g1=plot(x^3,x,-4,4, title='Here is a graph')
```

105

```
sage: g2=plot(x^2,x,-4,4, color= 'green',linestyle='--',
            thickness=2, legend_label='f(x)',figsize=3)
```

106

```
sage: g3=plot(2*x^2,x,-4,4, color= 'purple',figsize=3)
```

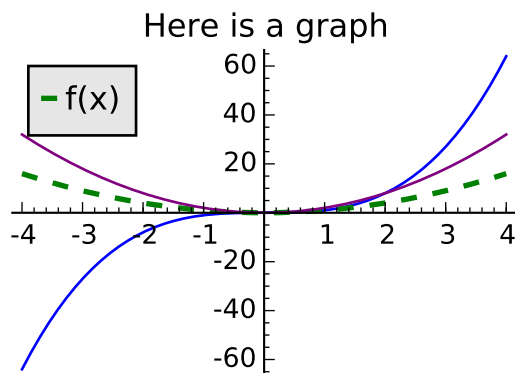
107

```
sage: g1+g2+g3
```

108

Graphics object consisting of 3 graphics primitives

109

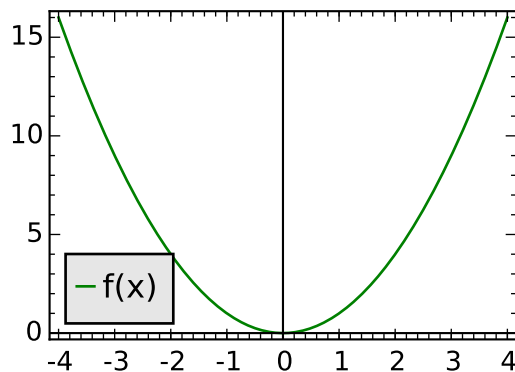


Example 2.1.12. Plot(x^2 , x , $-4, 4$) with color, frame, and label.

Solution:

```
sage: g=plot(x^2,x,-4,4, color= 'green',frame=True,
            legend_label='f(x)')
```

110

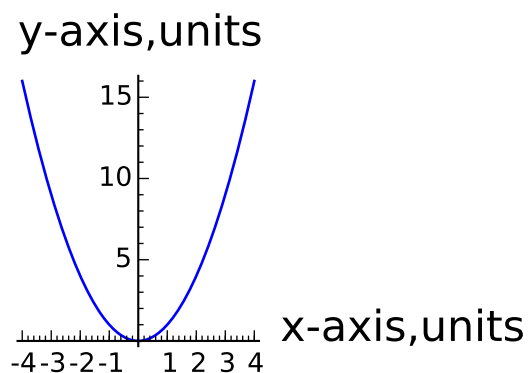


Example 2.1.13. Plot(x^2 , x , $-4, 4$) with axes.

Solution:

```
sage: g=plot(x^2,x,-4,4,axes_labels=['x-axis,units','y-axis,
            units'])
```

111



Example 2.1.14. Draw the ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 2$

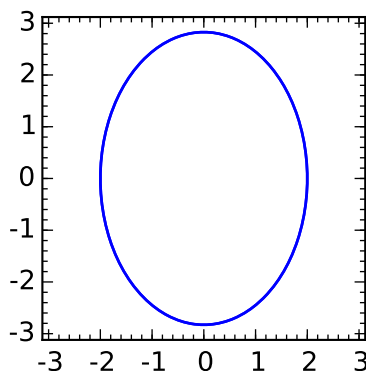
Solution:

```
sage: x,y=var('x,y')
```

112

```
sage: g=implicit_plot(x^2/2+y^2/4==2, (x, -3, 3), (y, -3,3))
```

113



2.2 Limits

2.2.1 Evaluating Limits

To compute the limit of function $f(x)$ as x approaches a :

$$\boxed{\text{limit}(f(x), x=a)}$$

To compute the limit of function $f(x)$ as x approaches a from the left (meaning $x < a$):

`limit(f(x),x=a,dir='minus')`

To compute the limit of function $f(x)$ as x approaches a from the right (meaning $x > a$):

`limit(f(x),x=a,dir='plus')`

Example 2.2.1. Evaluate $\lim_{x \rightarrow 1} \frac{2x^2+x+4}{x+1}$

Solution:

Following are tables of values of the function $\lim_{x \rightarrow 1} \frac{2x^2+x+4}{x+1}$ when x is sufficiently close to 1.

From the left:

<code>sage: def f(x): return (2*x^2+x+4)/(x+1)</code>	114
<code>sage: step=float(1/100)</code>	115
<code>sage: initial=float(9/10)</code>	116
<code>sage: table([(i*step+initial,f(i*step+initial)) for i in</code>	117
<code> [1..10]])</code>	
0.91 3.43780104712	118
0.92 3.44416666667	119
0.93 3.45067357513	120
0.94 3.45731958763	121
0.95 3.4641025641	122
0.96 3.47102040816	123
0.97 3.47807106599	124
0.98 3.48525252525	125
0.99 3.49256281407	126
1.0 3.5	127

From the right:

<code>sage: def f(x): return (2*x^2+x+4)/(x+1)</code>	128
<code>sage: step=float(-1/100)</code>	129

```

sage: initial=float(11/10)                                     130
sage: table([(i*step+initial,f(i*step+initial)) for i in      131
            [1..10]]))
1.09      3.57234449761                                         132
1.08      3.56384615385                                         133
1.07      3.5554589372                                         134
1.06      3.54718446602                                         135
1.05      3.53902439024                                         136
1.04      3.53098039216                                         137
1.03      3.52305418719                                         138
1.02      3.51524752475                                         139
1.01      3.50756218905                                         140
1.0       3.5                                                  141

```

From these tables, it is reasonable to expect that the limit is 3.5. Evaluating the limit confirm this:

```

sage: limit((2*x^2+x+4)/(x+1), x=1)                             142
7/2                                                                143

```

Example 2.2.2. Evaluate $\lim_{x \rightarrow 1} \frac{2x^2+x-1}{x+1}$

Solution:

```

sage: limit((2*x^2+x-1)/(x+1), x=1)                             144
1                                                                145

```

Example 2.2.3. Evaluate $\lim_{x \rightarrow 1^-} \frac{x^2-1}{x-1}$

Solution:

```
sage: limit((x^2-1)/(x-1), x=1, dir='minus')
```

146

```
2
```

147

Example 2.2.4. Evaluate $\lim_{x \rightarrow 1^+} \frac{x^2-1}{x-1}$

Solution:

```
sage: limit((x^2-1)/(x-1), x=1, dir='plus')
```

148

```
2
```

149

Example 2.2.5. Evaluate $\lim_{x \rightarrow -3} \frac{x+1}{x+3}$

Solution:

```
sage: limit((x+1)/(x+3), x=-3)
```

150

```
Infinity
```

151

Example 2.2.6. Show that $f(x) = 2 * \cos(1/x)$ does not have a limiting value as x approach 0.

Solution:

We define

```
sage: f(x)=2*cos(1/x)
```

152

```
sage: initial=float(1/10)
```

153

```
sage: step=float(-1/100)
```

154

```
sage: table([(i*step+initial, f(i*step+initial)) for i in
[1..8]])
```

155

```
0.09      0.230559899091497
```

156

```
0.08      1.99559655835716
```

157

```
0.07      -0.296003263241933
```

158

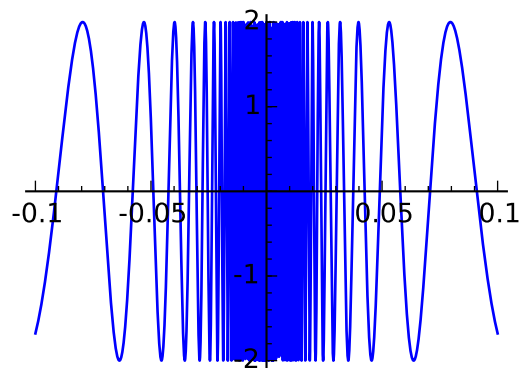
```
0.06      -1.14916333703824
```

159

0.05	0.816164123626784	160
0.04	1.98240562372695	161
0.03	-0.679423624807142	162
0.02	1.92993205698422	163

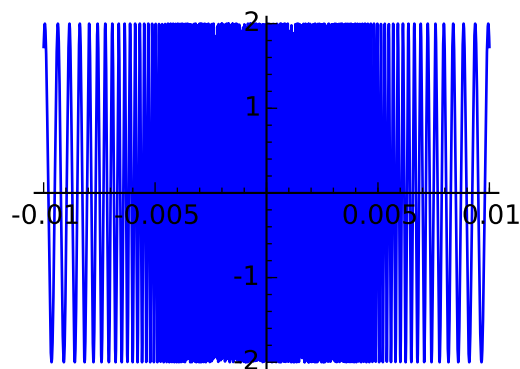
These values suggest that the limit does not exist. To make this clear, we consider the graph:

```
sage: g=plot(f(x),x,-1/10,1/10,figsize=3)
```

164


This indicates that there are too many oscillations around $x = 0$. Let us try to zoom in around this point:

```
sage: g=plot(f(x),x,-1/100,1/100,figsize=3)
```

165


Note that zooming in on this graph does not help. This indicates that the limit does not exist.

Example 2.2.7. Investigate the function $f(x) = \frac{x}{|x|}$ as $x \rightarrow 0$.

Solution:

```
sage: limit(x/abs(x), x=0, dir='minus')
```

166

```
-1
```

167

```
sage: limit(x/abs(x), x=0, dir='plus')
```

168

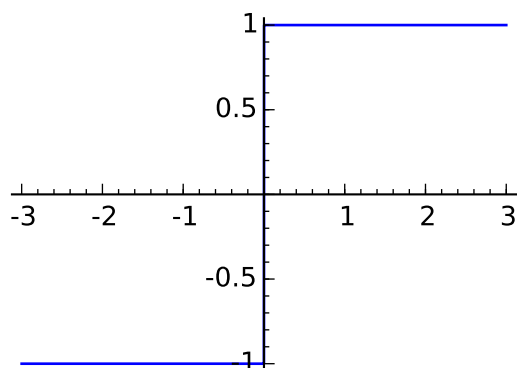
```
1
```

169

Since the left-hand and right-hand limits are not the same, we conclude that the limit does not exist.

```
sage: g=plot(x/abs(x), x, -3, 3, figsize=3)
```

170



2.2.2 Limits Involving Trigonometric Functions

For trigonometric functions, Sage uses the same traditional notation in calculus.

Example 2.2.8. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$

Solution:

```
sage: limit((sin(3*x)/x), x=0)
```

171

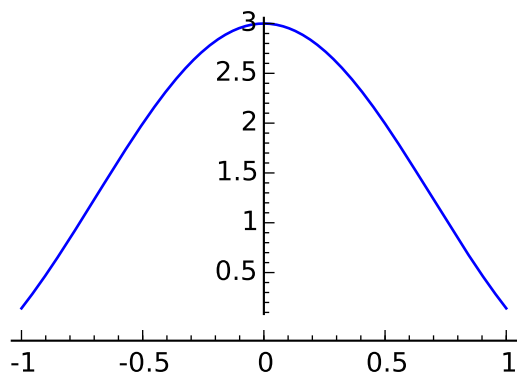
```
3
```

172

We can check the answer by graphing the function up close to the neighborhood of $x = 0$

```
sage: g=plot((sin(3*x)/x),x,-1,1)
```

173



Example 2.2.9. Evaluate $\lim_{t \rightarrow 0} \frac{\tan t}{|t|}$

Solution:

```
sage: limit((tan(x))/abs(x),x=0,dir='plus')
```

174

1

175

```
sage: limit((tan(x))/abs(x),x=0,dir='minus')
```

176

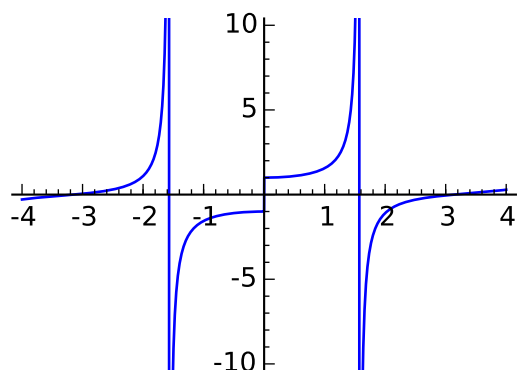
-1

177

Thus the limit does not exist. This can be clearly seen from the graph of the function below.

```
sage: g=plot((tan(x))/abs(x),x,-4,4,ymin=-10,ymax=10)
```

178



Example 2.2.10. Find

(a) $\lim_{x \rightarrow \pi} \frac{1}{\cos x}$

(b) $\lim_{x \rightarrow -\frac{\pi}{2}} 3^{\sin x}$

Solution:

```
sage: limit(1/cos(x), x=pi)
```

179

```
-1
```

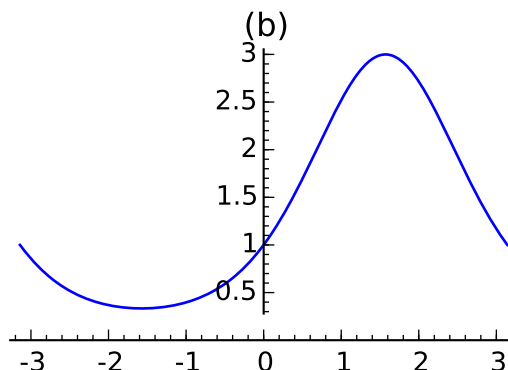
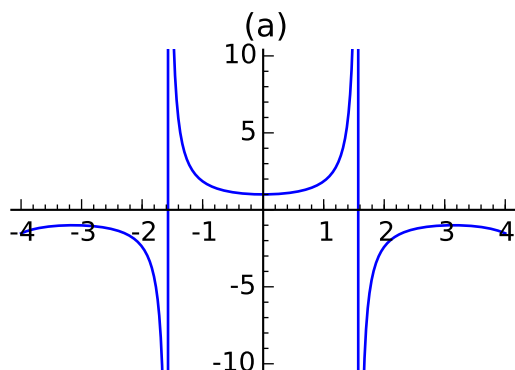
180

```
sage: limit(3^(sin (x)),x=- pi/2)
```

181

```
1/3
```

182



Example 2.2.11. Find $\lim_{x \rightarrow c} \frac{\sin x - \sin c}{\sin c}$ for values of $c = 0, \pi/6, \pi/4, \pi/3, \pi/2$.

Solution:

```
sage: c= [0,pi/6,pi/4,pi/3,pi/2]
```

```
sage: for i in range(5):
```

```
....:     limit((sin x - sin c[i])/(sin c[i]),x=c[i])
```

```
0, -1/2, -1/2*sqrt(2), -1/2*sqrt(3), -1
```

Example 2.2.12. Find $\lim_{x \rightarrow 0} \frac{\cos(nx) - 1}{x^2}$ for various values of n .

Solution:

Here is a table of limits for integer values of n ranging from 1 to 10. Notice that to avoid the confusing between an integer n and **n** command which returns numerical value, we always try to substitute integer n by i in sagecommandline:

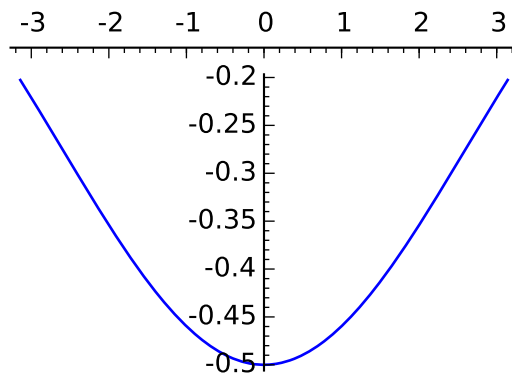
```
sage: table([(limit((cos(i*x)-1)/x^2,x=0))for i in [1..10]]) 183
      -1/2    -2    -9/2    -8    -25/2    -18    -49/2    -32    -81/2    184
      -50
```

A reasonable guess at a general formula for the answer would be $\lim_{x \rightarrow 0} (\cos(nx) - 1)/x^2 = -n^2/2$. We can check this with values of n ranging from 10 to 20.

```
sage: table([(limit((cos(i*x)-1)/x^2,x=0), -i^2/2)]for i in 185
      [10..20]))
      (-50, -50) 186
      (-121/2, -121/2) 187
      (-72, -72) 188
      (-169/2, -169/2) 189
      (-98, -98) 190
      (-225/2, -225/2) 191
      (-128, -128) 192
      (-289/2, -289/2) 193
      (-162, -162) 194
      (-361/2, -361/2) 195
      (-200, -200) 196
```

For a mathematical proof, first take $n = 1$ and plot the graph

```
sage: g=plot((cos(x)-1)/x^2,x,-pi,pi,figsize=3) 197
```



The graph above confirms that the limit is $-1/2$.

For the general case, let $t = nx$ so that $x^2 = \frac{t^2}{n^2}$. Then note that $x \rightarrow 0$ if and only if $t \rightarrow 0$. Thus, the limit can be evaluated in terms of t as

$$\lim_{x \rightarrow 0} \frac{\cos(nx) - 1}{x^2} = \lim_{t \rightarrow 0} \frac{\cos(t) - 1}{t^2/n^2} = n^2 \lim_{t \rightarrow 0} \frac{\cos(t) - 1}{t^2} = -\frac{n^2}{2}$$

2.2.3 Limits Involving Infinity

Example 2.2.13. Evaluate $\lim_{x \rightarrow \infty} \frac{3x-2}{\sqrt{x^2+2}}$ and $\lim_{x \rightarrow -\infty} \frac{3x-2}{\sqrt{x^2+2}}$

Solution:

```
sage: limit((3*x-2)/sqrt(x^2+2), x=infinity) 198
```

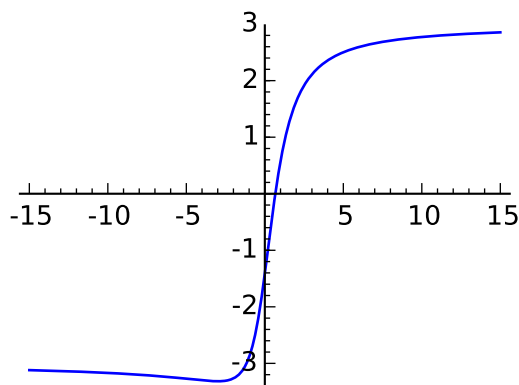
```
3 199
```

```
sage: limit((3*x-2)/sqrt(x^2+2), x=-infinity) 200
```

```
-3 201
```

Observe how the two limits differ. The following graph confirms this.

```
sage: a=plot((3*x-2)/sqrt(x^2+2), x,-15,15,figsize=3) 202
```



Example 2.2.14. Evaluate $\lim_{x \rightarrow 3^-} \frac{\sqrt{9-x^2}}{x-3}$

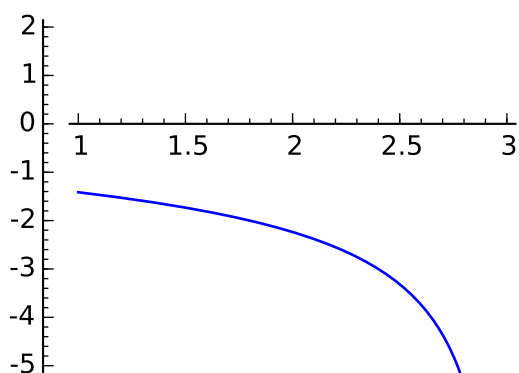
Solution:

```
sage: limit(sqrt(9-x^2)/(x-3), x=3, dir='minus') 203
```

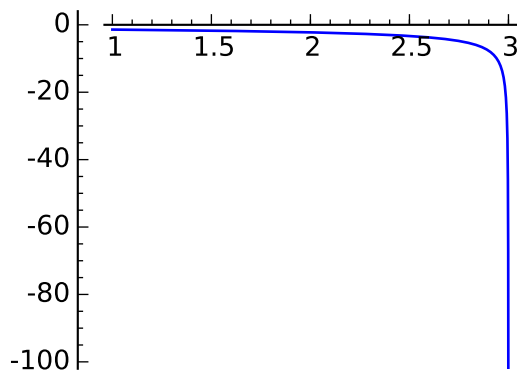
```
-Infinity 204
```

We plot the function over two different ranges to visually understand why the answer is $-\infty$. Notice how the first range fails to show this.

```
sage: g1=plot(sqrt(9-x^2)/(x-3), x,1,3,ymin=-5,ymax=2,figsize 205
      =3)
```



```
sage: g2=plot(sqrt(9-x^2)/(x-3), x,1,3,ymin=-100,ymax=2, 206
      figsize=3)
```



Example 2.2.15. Evaluate $\lim_{x \rightarrow \infty} \cos(x)$

Solution:

```
sage: limit(cos(x), x=infinity) 207
ind 208
```

Example 2.2.16. Find $\lim_{x \rightarrow \infty} \frac{\cos x}{x}$

Solution:

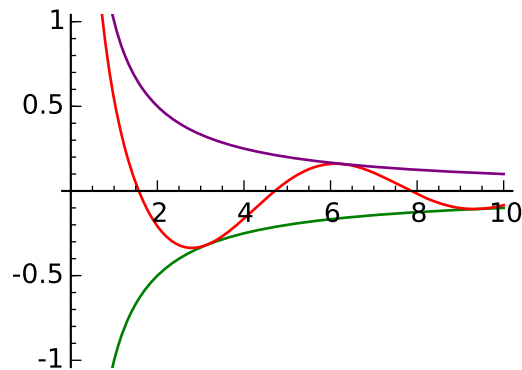
```
sage: limit(cos(x)/x, x=infinity) 209
0 210
```

We can verify this limit by using the Squeeze Theorem. In this case, we take $f(x) = -\frac{1}{|x|}$, $g(x) = \frac{\cos x}{x}$, and $h(x) = \frac{1}{|x|}$. Then $f(x) \leq g(x) \leq h(x)$ since $-1 \leq \cos x \leq 1$

```
sage: g1=plot((-1/abs(x)),x,0,10,ymin=-1,ymax=1,color='green', 211
        figsize=3)
sage: g2=plot(cos(x)/x,x,0,10,ymin=-1,ymax=1,color='red', 212
        figsize=3)
sage: g3=plot(1/abs(x),x,0,10,ymin=-1,ymax=1,color='purple', 213
        figsize=3)
```

```
sage: g=g1+g2+g3
```

214



Since $-\frac{1}{|x|}$ and $\frac{1}{|x|}$ both approach 0 as $x \rightarrow \infty$, we conclude that $\frac{\cos x}{x}$ approaches zero as well.

Example 2.2.17. Evaluate $\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$

Solution:

```
sage: limit(1/log(x)-1/(x-1), x=1, dir='plus')
```

215

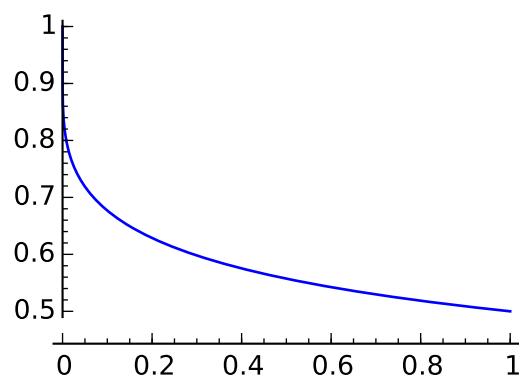
```
1/2
```

216

We can graph the function near $x = 1$ to visually understand why the answer is $1/2$:

```
sage: g=plot(1/log(x)-1/(x-1), x,0,1, figsize=3)
```

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Note, however, that this example shows that $1/\ln x$ and $1/(x-1)$ both grow to ∞ at the same rate as $x \rightarrow 1^+$

Example 2.2.18. Let $f(x) = \frac{x^{2n}-1}{x^{2m}-1}$. Evaluate $\lim_{x \rightarrow 1} f(x)$ by substituting in various values of m and n .

Solution:

```
sage: table([[limit((x^i-1)/(x^j-1),x=1) for i in [1..8]] for j in [1..8]], align='center',frame=True, header_row=['i_1','i_2','i_3','i_4','i_5','i_6','i_7','i_8'], header_column=['j_1','j_2','j_3','j_4','j_5','j_6','j_7','j_8'])
```

	j_1	j_2	j_3	j_4	j_5	j_6	j_7	j_8
i_1	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	$\frac{1}{8}$
i_2	2	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
i_3	3	$\frac{3}{2}$	1	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{2}{3}$	$\frac{3}{7}$	$\frac{2}{5}$
i_4	4	2	$\frac{4}{3}$	1	$\frac{4}{5}$	$\frac{3}{2}$	$\frac{4}{7}$	$\frac{3}{4}$
i_5	5	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	1	$\frac{5}{6}$	$\frac{5}{7}$	$\frac{4}{5}$
i_6	6	3	2	$\frac{6}{4}$	$\frac{6}{5}$	1	$\frac{6}{7}$	$\frac{5}{6}$
i_7	7	$\frac{7}{2}$	$\frac{7}{3}$	$\frac{7}{4}$	$\frac{7}{5}$	$\frac{7}{6}$	1	$\frac{7}{8}$
i_8	8	4	$\frac{8}{3}$	2	$\frac{8}{5}$	$\frac{8}{6}$	$\frac{8}{7}$	1

Can you guess a formula for $\lim_{x \rightarrow 1} f(x)$ in term of m and n ? Enter the command **limit**(($x^n - 1$)/($x^m - 1$), $x = 1$) into an input cell and evaluate it to verify your conjecture.

Let us end this section with an example where the **limit** command is used to evaluate the derivative of a function (in anticipation of commands introduced in the next chapter for computing derivatives).

By definition, the derivative of a function f at x (i.e., the slope of its tangent line at x) is

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Example 2.2.19. Find the derivative of $f(x) = \frac{1}{4x}$ according to the limit definition.

Solution:

We first exam the derivative by tabulating values of the difference quotient, $\frac{f(x+\Delta x)-f(x)}{\Delta x}$, for some arbitrarily chosen values of Δx :

```
sage: f(x)= 1/(4*x) 238
sage: var('c') 239
c 240
sage: c=[10^(-1), 10^(-2), 10^(-4), 10^(-5), 10^(-6), 10^(-8)] 241
sage: table([(n(c[i],digits=4),((f(x+c[i]))-f(x))/c[i])) for i 242
in [0..5]])
0.1000      25/(10*x + 1) - 5/2/x 243
0.01000     2500/(100*x + 1) - 25/x 244
0.0001000   25000000/(10000*x + 1) - 2500/x 245
0.00001000  2500000000/(100000*x + 1) - 25000/x 246
1.000e-6     250000000000/(1000000*x + 1) - 250000/x 247
1.000e-8     2500000000000000/(100000000*x + 1) - 25000000/x 248
```

This table suggest that $f'(x) = -1/(4x^2)$ in the limit as $\Delta x \rightarrow 0$. We confirm this with Sage:

```
sage: limit((f(x+Deltax)-f(x))/Deltax ,Deltax=0) 249
```

$$-1/4/x^2$$

250

2.3 Continuity

Recall that a function is continuous at $x = a$ if and only if $\lim_{x \rightarrow a} f(x) = f(a)$. Graphically, this means that there is no break (or jump) in the graph of f at the point $(a, f(a))$. It is not possible to indicate this discontinuity using computer graphics for the situation where the limit exists and the function is defined at a but the limit is not equal to $f(a)$. For other cases of discontinuity, computer graphics are very helpful.

To verify if a given function is continuous at a point, we evaluate its limit there and check if this limit is equal to the value of the function.

Example 2.3.1. Show that the function $f(x) = x^3 - 1$ is continuous everywhere.

Solution:

We could draw the graph and observe this fact. On the other hand, we can get Sage to check continuity:

```
sage: def f(x): return x^3-1          251
sage: var('c')                        252
c                                     253
sage: bool(limit(f(x), x=c)==f(c))   254
True                                 255
```

This means that $\lim_{x \rightarrow c} f(x) = f(c)$ and hence f is continuous everywhere.

Example 2.3.2. Find point of discontinuity for each of the following function:

$$(a) \text{ Let } f(x) = \begin{cases} \frac{x^2-1}{x-1}, & \text{if } x \neq 1 \\ 2, & \text{if } x = 1 \end{cases}$$

$$(b) \text{ Let } g(x) = \begin{cases} \frac{x^2-1}{x-1}, & \text{if } x \neq 1 \\ 6, & \text{if } x = 1 \end{cases}$$

Solution:

The piece-wise functions can be defined by using **if, else**:

(a) Define the function f:

```
sage: def f(x):
.....:     if x<>1:
.....:         return (x^2-1)/(x-1)
.....:     else:
.....:         return 2
```

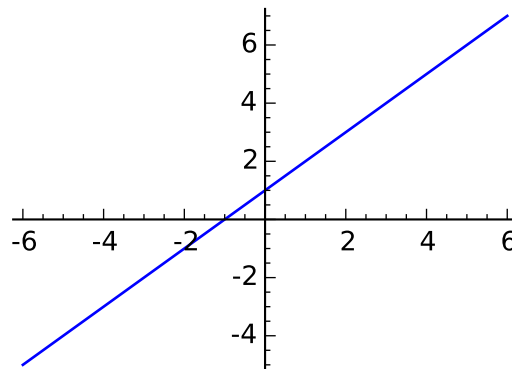
Then we can check continuity of f at $x = 1$.

```
sage: bool(limit(f(x),x=1)==f(1))
True
```

Hence, the function is continuous at $x = 1$. For continuity at other points, we observe that the rational function $\frac{x^2-1}{x-1}$ simplifies to $x+1$ in this case (factor the numerator!) and thus is continuous at any point except $x = 1$. Thus, f is continuous everywhere. We can also confirm this by examining the graph of f below.

```
sage: g=plot(f(x),x,-6,6,figsize=3)
```

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(b) As in part a, we define the function and consider continuity of g at $x = 1$:

```

sage: def g(x):
...:     if x<>1:
...:         return (x^2-1)/(x-1)
...:     else:
...:         return 6

```

```

sage: bool(limit(g(x),x=1)==g(1))
False

```

Thus, g is NOT continuous at $x = 1$. For continuity at other point, we again observe that the rational function $\frac{x^2-1}{x-1} = x+1$ and thus is continuous for $x \neq 1$.

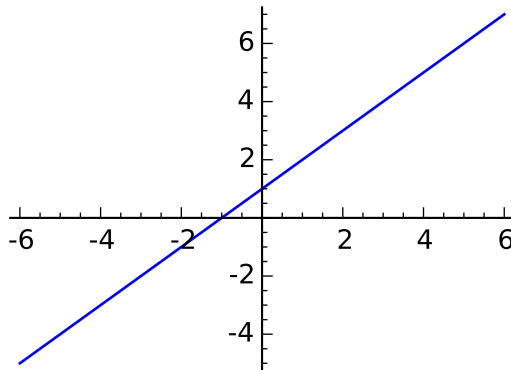
Caution: The plot of the graph of g given below indicates (incorrectly) that g is continuous everywhere! Care must be taken when examining Sage plots to draw conclusion about continuity.

```

sage: g=plot(g(x),x,-6,6,figsize=3)

```

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Example 2.3.3. Let $f(x) = \begin{cases} \cos(\frac{1}{x}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$. Prove that for any number k between -1 and 1 there exists a value for c such that $f(c) = k$.

Solution:

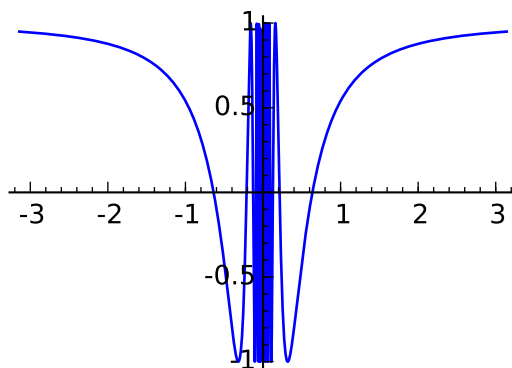
Note: observe that f is not continuous at $x = 0$ so the converse of the Intermediate Value Theorem does not hold.

For $k = 0$, we choose $c = 0$ so that $f(c) = 0$. For any nonzero k between -1 and 1 , define $y =$

$\cos^{-1}k$ (using the principal domain of the $\cos$ function) and let $c = 1/y$. Then $f(c) = \cos(1/c) = \cos y = k$. The graph of f following shows that there are in fact infinitely many choices for c .

```
sage: g=plot(cos(1/x),x,-pi,pi,figsize=3)
```

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Chapter 3

Differentiation

3.1 The Derivative

In this section, we introduce few more popular commands in Sage.

- To calculate the derivative of a function, use **diff()** or **.derivative()** command:

diff(f(x)) or f(x).derivative()

- To differentiate $f(x, y)$ with respects to x :

diff(f(x,y),x)

- To compute the n derivative respect to x :

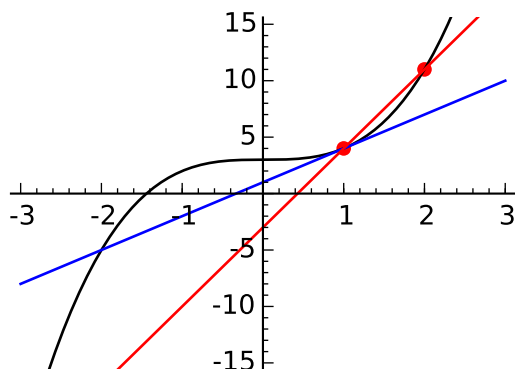
diff(f(x),x,n)

3.1.1 Slope of Tangent

The most fundamental concepts in calculus is the derivative. Its definition is given by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(h+a) - f(a)}{h}$$

where geometrically $f'(a)$ is the slope of the line tangent to the graph of $f(x)$ at $x = a$, provided that the limit exists. We can view this graphically in the illustration below, where the tangent line (shown in blue) is viewed as a limit of secant lines (one shown in red) as $h \rightarrow 0$.



Example 3.1.1. Calculate the derivative of $f(x) = \frac{x^4}{3}$ at $x = 1$ using the point-wise definition of a derivative.

Solution:

We first use the **table** command to tabulate slopes of secant lines passing through the points at $a = 1$ and $a + h = 1 + h$ by choosing arbitrarily small values for h (taken as reciprocal powers of 10)

```
sage: a,x,i=var('a,x,i') 259
```

```
sage: f(x)=x^4/3 260
```

```
sage: a=1 261
```

```
sage: table([(n(1/(10^i),digits=4), n((f(a+1/(10^i))-f(a)) 262
            /(1/(10^i)),digits=4)) for i in [1..5]])
```

```
0.1000      1.547 263
```

```
0.01000     1.353 264
```

```
0.001000    1.335 265
```

```
0.0001000   1.334 266
```

```
0.00001000  1.333 267
```

Note that our use of the **table** command, which displays a list as an array of rectangular cells. From the table output, we may conclude that $f'(1) = 4/3$. A more rigorous approach is to algebraically simplify the difference quotient, $\frac{f(a+h)-f(a)}{h}$

```
sage: ((f(a+h)-f(a))/h).simplify_full()
```

```
1/3*h^3 + 4/3*h^2 + 2*h + 4/3
```

It is now clear that $\frac{f(a+h)-f(a)}{h} \rightarrow \frac{4}{3}$ as $h \rightarrow 0$. This can be checked using Sage **limit** command:

```
sage: limit((f(a+h)-f(a))/h,h=0)
```

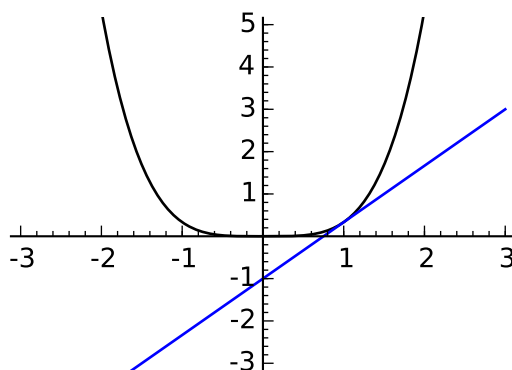
```
4/3
```

Below is a plot of the graph of $f(x)$ (in black) and its corresponding tangent line (in blue), which also confirms our answer:

```
sage: g1=plot(f(x),x,-3,3,ymin=-3,ymax=5,figsize=3,color='black')
```

```
sage: ff(x)=diff(f(x))
```

```
sage: g2=plot(ff(a)*(x-a)+f(a),x,-3,3,ymin=-3,ymax=5,figsize=3)
```



Recall that the tangent line of $f(x)$ at $x = a$ is given by:

$$y = f'(a)(x - a) + f(a)$$

3.1.2 Derivative as a Function

The derivative is best represented of as a slope function, one that gives the slope of the tangent line at any point on the graph of $f(x)$ where this slope exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example 3.1.2. Compute the derivative of $\cos(x^2)$ and evaluate it at $x = \sqrt{\pi/4}$

Solution:

```
sage: f(x)=cos(x^2) 275
sage: diff(f(x)).substitute(x=sqrt(pi/4)) 276
-1/2*sqrt(2)*sqrt(pi) 277
```

where **substitute()** command inserts values of variable in () into function $f'(x)$.

Note: Observe that the derivative of $\cos(x^2)$ is NOT $-\sin(x^2)$ but $-2x\sin(x^2)$. This is because $\cos(x^2)$ is a composite function. It's a rule for differentiating composite functions, known as the Chain Rules.

Example 3.1.3. Compute the derivative of $f(x) = \begin{cases} \frac{\cos x}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Solution:

To define functions described by two different formulas over separate domains, we employ Sage

Piecewise command.

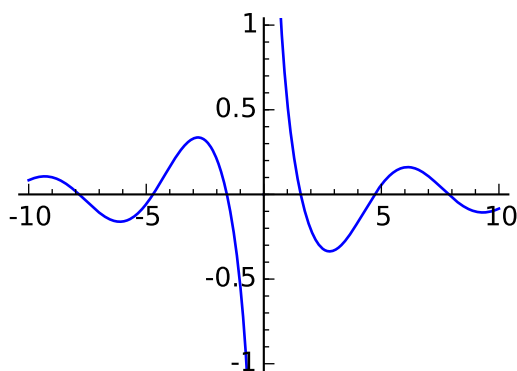
```
sage: f1(x)=cos(x)/x 278
sage: f2(x)=0 279
sage: f = Piecewise([[ (0,0),f2],[(-infinity,0),f1],[ (0, 280
infinity),f1]] )
sage: f.derivative() 281
```

Piecewise defined function with 3 parts, $[(0, 0), x \mapsto 0]$, 282
 $[(-\infty, 0), x \mapsto -\sin(x)/x - \cos(x)/x^2]$, $[(0, +$
 $\infty), x \mapsto -\sin(x)/x - \cos(x)/x^2]$

Note: It is clear for $x \neq 0$ that the derivative is $-\frac{\sin(x)}{x} - \frac{\cos(x)}{x^2}$ as a result of the Quotient Rule .

Notice that the fact that $f(0) = 0$ does not mean that f is a constant.

A plot of the graph of $f(x)$ reveals that it is discontinuous at $x = 0$, and thus not differentiable there:



Example 3.1.4. Find the equation of the tangent line to the graph of $f(x) = \sqrt{2x+2}$ at $x = 2$.

Solution:

Remember that the tangent line to a function $f(x)$ at $x = a$ is $L(x) = f(a) + f'(a)(x - a)$. Here, $a = 2$:

```
sage: f(x)=sqrt(2*x+2) 283
```

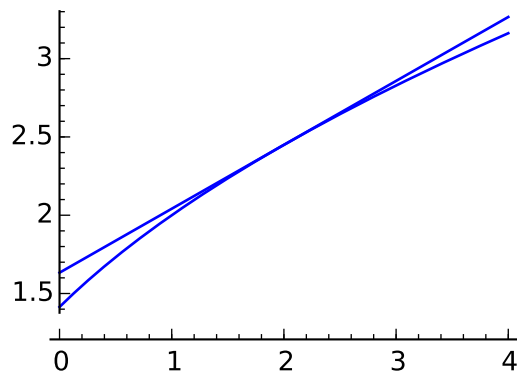
```
sage: L(x)=f(2)+diff(f(x)).substitute(x=2)*(x-2) 284
```

```
sage: L(x) 285
```

```
1/6*sqrt(6)*(x - 2) + sqrt(6) 286
```

To see that $L(x)$ is indeed the desired tangent line, we will plot f and L together:

```
sage: g=plot((f(x),L(x)),x,0,4,figsize=3) 287
```



Example 3.1.5. Find an equation of the line passing through the point $P(2, -3)$ and tangent to the graph of $f(x) = x^2 + 1$

Solution:

Let us refer to $Q(a, f(a))$ as the point of tangency for our desired tangent line. To determine Q , we compute the slope of our desired tangent line from two different perspectives:

Slope of line segment PQ:

```
sage: var('a') 288
a 289
sage: f(x)=x^2+1 290
sage: m=(f(a)-(-3))/(a-2) 291
sage: m 292
(a^2 + 4)/(a - 2) 293
```

Derivative of $f(x)$ at $x = a$:

```
sage: f(x)=x^2+1 294
sage: diff(f(x)).substitute(x=a) 295
2*a 296
```

Equating the two formulas for slope above and solving for a yields:

```
sage: solve(m==diff(f(x)).substitute(x=a), a) 297
```

```

[                                                                    298
a == -2*sqrt(2) + 2,                                                299
a == 2*sqrt(2) + 2                                                  300
]                                                                    301

```

Since there are two valid solutions for a , we have in fact found two such tangent lines. Their equations are given by:

$$y_1 = f'(a)(x - a) + f(a) \text{ as } a \rightarrow 2 - 2\sqrt{2}$$

$$y_2 = f'(a)(x - a) + f(a) \text{ as } a \rightarrow 2 + 2\sqrt{2}$$

```

sage: y1(x)=(diff(f(x)).substitute(x=a)*(x-a)+f(x).substitute( 302
      x=a)).substitute(a=2*sqrt(2)+2).simplify_full()
sage: y1(x)                                                         303
4*x*(sqrt(2) + 1) - 8*sqrt(2) - 11                                  304
sage: y2(x)=(diff(f(x)).substitute(x=a)*(x-a)+f(x).substitute( 305
      x=a)).substitute(a=-2*sqrt(2)+2).simplify_full()
sage: y2(x)                                                         306
-4*x*(sqrt(2) - 1) + 8*sqrt(2) - 11                                307

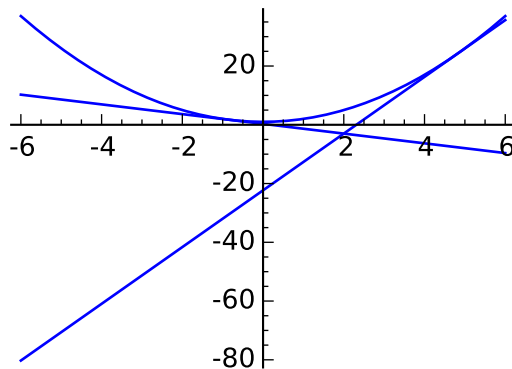
```

Plotting these tangent lines together with the graph of $f(x)$ confirms that our solution is correct:

```

sage: g=plot((f(x),y1(x),y2(x)),x,-6,6,figsize=3)                  308

```



3.2 Higher-Order Derivatives

Suppose we are interested in pursuing higher order derivatives of a function. The reasons are they relate to applications of minimum and maximum values, physical applications such as velocity and acceleration, or finding points of inflection.

Example 3.2.1. Compute the first eight derivatives of $f(x) = \cos(x)$. What is the 255th derivative of f ?

Solution:

Here are the first eight derivative of f :

```
sage: f(x)=cos(x) 309
sage: ([diff(f(x),x,i) for i in [1..8]]) 310
[-sin(x), -cos(x), sin(x), cos(x), -sin(x), -cos(x), sin(x), 311
cos(x)]
```

We observe from the output that the higher-order derivatives of f are periodic modulo 4, which means they repeat every four derivative. Since 255 has remainder 3 divided by 4, it follows that

$$f^{(255)}(x) = f^{(3)}(x) = \sin(x)$$

Of course, Sage can compute this derivative (see output below), but the pattern above gives us a

more in-depth understanding of the higher-order derivatives of $\cos(x)$.

```
sage: diff(f(x), x, 255) 312
```

```
sin(x) 313
```

Example 3.2.2. Compute the first three derivatives of $f(x) = x\sin(x)$

Solution:

We use the command **diff(f(x),x,n)** to compute the n^{th} derivative of f . Here, we set $n = 1, 2, 3$

```
sage: f(x)=x*sin(x) 314
```

```
sage: diff(f(x), x) 315
```

```
x*cos(x) + sin(x) 316
```

```
sage: diff(f(x), x, 2) 317
```

```
-x*sin(x) + 2*cos(x) 318
```

```
sage: diff(f(x), x, 3) 319
```

```
-x*cos(x) - 3*sin(x) 320
```

A quicker way to generate a list of higher-order derivatives is to use the **table** command. For example, here is a list of the first five derivatives of f :

```
sage: ([diff(f(x), x, i) for i in [1..5]]) 321
```

```
[x*cos(x) + sin(x), -x*sin(x) + 2*cos(x), -x*cos(x) - 3*sin(x) 322  
 , x*sin(x) - 4*cos(x), x*cos(x) + 5*sin(x)]
```

3.3 Chain Rule and Implicit Differentiation

In this section, we demonstrate not only how Sage uses the Chain Rule to differentiate composite functions but also to compute derivatives of functions defined implicitly by equations where solving for the dependent variable is not desirable.

Example 3.3.1. Find all horizontal tangents of $f(x) = \sqrt{\frac{2x^4-2x+1}{2x^4+x+1}}$

Solution:

We first compute the derivative of f , which requires the Chain Rule.

```
sage: f(x)=sqrt((2*x^4-2*x+1)/(2*x^4+x+1)) 323
```

```
sage: diff(f(x)).simplify_full() 324
```

```
3/2*(6*x^4 - 1)/((4*x^8 + 4*x^5 + 4*x^4 + x^2 + 2*x + 1)*sqrt 325
((2*x^4 - 2*x + 1)/(2*x^4 + x + 1)))
```

Horizontal tangents have zero slope and so it suffices to solve $f'(x) = 0$ for x .

```
sage: solve(diff(f(x))==0,x) 326
```

```
[ 327
```

```
x == 1/6*I*6^(3/4), 328
```

```
x == -1/6*6^(3/4), 329
```

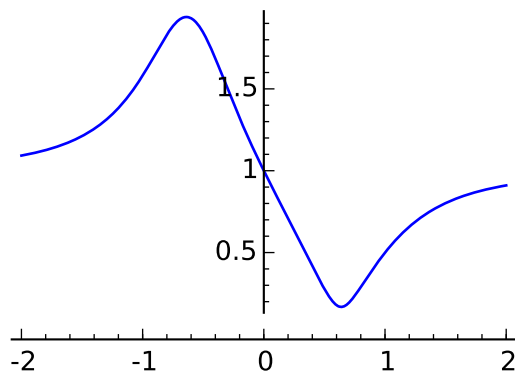
```
x == -1/6*I*6^(3/4), 330
```

```
x == 1/6*6^(3/4) 331
```

```
] 332
```

Observe that the solutions above are nothing more than the zeros of the numerator of $f'(x)$. We ignore the first and third solutions listed above, which are imaginary. Hence, $x = \frac{1}{6} * 6^{\frac{3}{4}} = 0.6389$ and $x = -\frac{1}{6} * 6^{\frac{3}{4}} = -0.6389$. A plot of the graph of f below confirms our solution.

```
sage: g=plot(f(x),x,-2,2,figsize=3) 333
```



Example 3.3.2. Find all horizontal tangents of the lemniscate described by $4(x^2 + y^2)^2 = 15(x^2 - y^2)$

Solution:

Implicit differentiation is required here to compute $\frac{dy}{dx}$, which involves first differentiating the lemniscate equation and then solving for our derivative. Observe that we make the substitution $y \rightarrow y(x)$, which makes explicit our assumption that y depends on x .

```
sage: var('x,y') 334
(x, y) 335
sage: y(x)=function('y')(x) 336
sage: eq=4*(x^2+y^2)^2==15*(x^2-y^2) 337
sage: eq.substitute(y=y(x)) 338
x |--> 4*(x^2 + y(x)^2)^2 == 15*x^2 - 15*y(x)^2 339
sage: diff(eq,x) 340
x |--> 16*(x^2 + y(x)^2)*(y(x)*D[0](y)(x) + x) == -30*y(x)*D 341
      [0](y)(x) + 30*x
sage: solve(diff(eq,x),diff(y(x))) 342
[ 343
D[0](y)(x) == -(8*x^3 + 8*x*y(x)^2 - 15*x)/(8*y(x)^3 + (8*x^2 344
      + 15)*y(x))
] 345
```

Notice that $D[0](y)(x)$ is the first derivative of $y(x)$ or $y'(x)$.

To find horizontals, it suffices to find where the numerator of $y'(x)$ vanishes (since the denominator never vanishes except when $y = 0$). Thus, we solve the system of equations

$$\begin{cases} 15x - 8x^3 - 8xy^2 = 0 \\ 4(x^2 + y^2)^2 = 15(x^2 - y^2) \end{cases}$$

since the solutions must also lie on the lemniscate.

```
sage: var('x,y') 346
(x, y) 347
sage: solve([4*(x^2+y^2)^2==15*(x^2-y^2), 15*x-8*x^3-8*x*y 348
^2==0], x,y)
[ 349
[x == 0, y == -1/2*I*sqrt(15)], 350
[x == 0, y == 1/2*I*sqrt(15)], 351
[x == 0, y == 0], 352
[x == -3/8*sqrt(5)*sqrt(2), y == -1/8*sqrt(15)*sqrt(2)], 353
[x == -3/8*sqrt(5)*sqrt(2), y == 1/8*sqrt(15)*sqrt(2)], 354
[x == 3/8*sqrt(5)*sqrt(2), y == -1/8*sqrt(15)*sqrt(2)], 355
[x == 3/8*sqrt(5)*sqrt(2), y == 1/8*sqrt(15)*sqrt(2)] 356
] 357
```

From the output, we see that the last four solutions are valid:

$$(-3/8 * \sqrt{5} * \sqrt{2}, -1/8 * \sqrt{15} * \sqrt{2}) \approx (-1.186, -0.685),$$

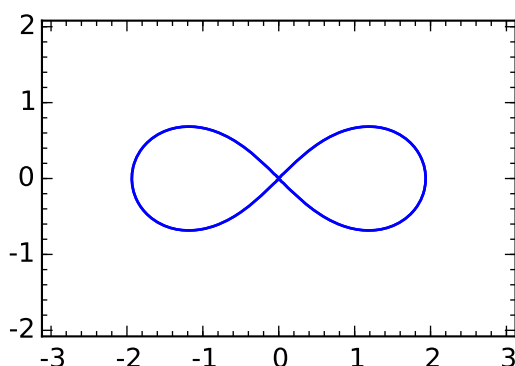
$$(-3/8 * \sqrt{5} * \sqrt{2}, 1/8 * \sqrt{15} * \sqrt{2}),$$

$$(3/8 * \sqrt{5} * \sqrt{2}, -1/8 * \sqrt{15} * \sqrt{2}),$$

$$(3/8 * \sqrt{5} * \sqrt{2}, 1/8 * \sqrt{15} * \sqrt{2})$$

which can be confirmed by inspecting the graph of the lemniscate below. Observe the symmetry in the solutions.

```
sage: x,y=var('x,y') 358
sage: g=implicit_plot(4*(x^2+y^2)^2==15*(x^2-y^2), (x,-3,3), (y 359
, -2,2), figsize=3)
```



3.4 Derivatives of Inverse, Exponential and Logarithmic Functions

3.4.1 Inverse Function

Recall that a function $g(x)$ is the inverse of a given function $f(x)$ if $f(g(x)) = g(f(x)) = x$. The inverse of $f(x)$ is denoted by $f^{-1}(x)$. We note that a necessary and sufficient condition for a function to have an inverse is that it must be one-to-one. On the other hand, a function is one-to-one if it is strictly increasing or strictly decreasing throughout its domain.

Example 3.4.1. Determine if the function $f(x) = 2x^2 - 2x + 1$ has an inverse on the domain $(-\infty, \infty)$. If it exists, then find the inverse.

Solution:

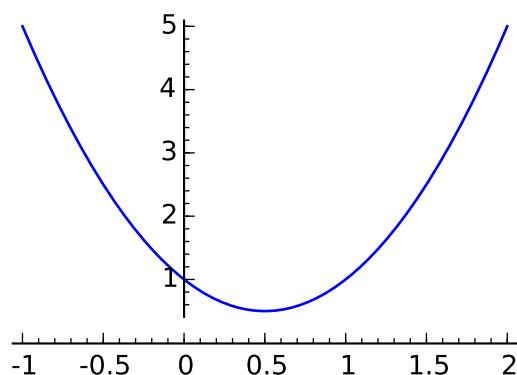
We note that $f(0) = f(1) = 1$. Thus, f is not one-to-one. We can also plot the graph of f and note that it fails the Horizontal Line Test since it is not increasing on its domain.

```
sage: f(x)=2*x^2-2*x+1
```

360

```
sage: g=plot(f(x),x,-1,2,figsize=3)
```

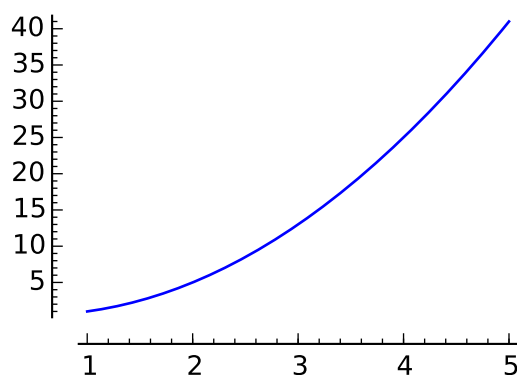
361



However, observe that if we restrict the domain of f to an interval where f is either increasing or decreasing, say $[1, \infty]$, then its inverse exists:

```
sage: g=plot(f(x),x,1,5,figsize=3)
```

362



To find the inverse on this restricted domain, let $y = f^{-1}(x)$. Then $f(y) = x$. Thus, we solve for y from the equation $f(y) = x$.

```
sage: var('x,y')
```

363

```
(x, y)
```

364

```

sage: sol=solve(f(y)==x,y) 365
sage: sol 366
[ 367
y == -1/2*sqrt(2*x - 1) + 1/2, 368
y == 1/2*sqrt(2*x - 1) + 1/2 369
] 370

```

Note that Sage gives two solutions. Only the second one is valid because it has range $[1, \infty]$, which agrees with the domain of f . Therefore,

$$f^{-1}(x) = \frac{1}{2}(1 + \sqrt{2x-1})$$

To extract this solution from the above output, we use the syntax below and denote the inverse function in Sage by $g(x)$.

```

sage: g(x)=sol[1].rhs() 371
sage: g(x) 372
1/2*sqrt(2*x - 1) + 1/2 373

```

Note: One can also attempt to verify $g(f(x)) = x$. However, Sage cannot confirm this identity:

```

sage: g(f(x)).simplify_full() 374
1/2*sqrt(4*x^2 - 4*x + 1) + 1/2 375

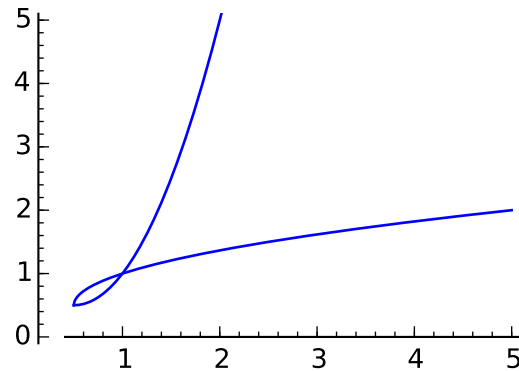
```

Lastly, a plot of the graph of $f(x)$ and $g(x)$ shows their expected symmetry about the diagonal line $y = x$.

```

sage: h=plot((f(x),g(x)),x,1/2,5,figsize=3,ymin=0,ymax=5) 376

```



Example 3.4.2. Determine if the function $f(x) = 2x^3 + 3x$ has an inverse. If it exists, then compute $(f^{-1})'(2)$.

Solution:

Since $f'(x) = 6x^2 + 3$, f is increasing on its domain therefore it has an inverse. Again, we can solve for this inverse as in the previous example:

```
sage: var('f,g,x,y,sol')
(f, g, x, y, sol)
sage: f(x)=2*x^3+3*x
sage: sol=solve(f(y)==x,y)
sage: sol
[
y == -1/2*(1/4*x + 1/4*sqrt(x^2 + 2))^(1/3)*(I*sqrt(3) + 1) +
1/4*(-I*sqrt(3) + 1)/(1/4*x + 1/4*sqrt(x^2 + 2))^(1/3),
y == -1/2*(1/4*x + 1/4*sqrt(x^2 + 2))^(1/3)*(-I*sqrt(3) + 1) +
1/4*(I*sqrt(3) + 1)/(1/4*x + 1/4*sqrt(x^2 + 2))^(1/3),
y == (1/4*x + 1/4*sqrt(x^2 + 2))^(1/3) - 1/2/(1/4*x + 1/4*sqrt
(x^2 + 2))^(1/3)
]
```

Only the third solution listed above is valid, being real valued. Thus:

$$f^{-1}(x) = \left(\frac{x}{4} + \frac{\sqrt{x^2+2}}{4}\right)^{\frac{1}{3}} - \frac{1}{2\left(\frac{x}{4} + \frac{\sqrt{x^2+2}}{4}\right)^{\frac{1}{3}}}$$

Denote our inverse as:

```
sage: g(x)=sol[2].rhs() 387
```

```
sage: g(x) 388
```

```
(1/4*x + 1/4*sqrt(x^2 + 2))^(1/3) - 1/2/(1/4*x + 1/4*sqrt(x^2 389
+ 2))^(1/3)
```

Lastly, we compute $g'(2)$:

```
sage: n(diff(g(x)).substitute(x=2), digits=3) 390
```

```
0.207 391
```

3.4.2 Exponential and Logarithmic Functions

One of the most important functions in mathematics and its applications is the exponential function.

In particular, the natural exponential function $f(x) = e^x$, where

$$e = \lim_{x \rightarrow 0} (1+x)^{1/x} \approx 2.718$$

In Sage, we use the lower letter e to denote the Euler number:

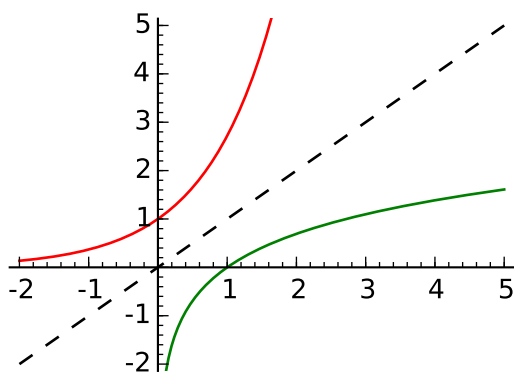
```
sage: limit((1+x)^(1/x), x=0) 392
```

```
e 393
```

Every exponential function $f(x) = a^x$, $a \neq 1$, $a > 0$, has domain $(-\infty, \infty)$ and range $(0, \infty)$. It is also one-to-one on its domain. Hence, it has an inverse. The inverse of an exponential function $f(x) = a^x$ is called the logarithm function and its denoted by $g(x) = \log_a x$. The inverse of the natural exponential function is denoted by $g(x) = \ln x$ and is called the natural logarithm. In Sage, we use $\log(x)$ for $\ln x$. Below is a plot of the graphs of e^x and $\ln x$ in red and green, respectively.

Observe their symmetry about the dashed line $y = x$.

```
sage: x=var('x') 394
sage: h1=plot(e^x,x,-2,5,figsize=3,color='red',ymin=-2,ymax=5) 395
sage: h2=plot(log(x),x,-2,5,figsize=3,color='green',ymin=-2, 396
ymax=5)
sage: h3=plot(x, x,-2,5,figsize=3,linestyle='--',color='black' 397
,ymin=-2,ymax=5)
```



Please refer to Section 3.9 of Rogawski's Calculus book for derivative formulas of general exponential and logarithmic functions.

Example 3.4.3. Compute derivative of the following functions.

(a) $f(x) = 2^x$ (b) $f(x) = 2x^2 + e^x$ (c) $f(x) = \ln x^3$

Solution:

We will input the functions directly and use the command **diff**. Note that $\log(x^3)$ should read as $\ln x^3$.

(a)

```
sage: diff(2^x) 398
```

```
2^x*log(2) 399
```

(b)

```
sage: diff(2*x^2+e^x) 400
```

```
4*x + e^x 401
```

(c)

```
sage: diff(log(x^3)) 402
```

```
3/x 403
```

Example 3.4.4. Find point on the graph of $f(x) = x^2e^{3x+5} + 3x$ where the tangent lines are parallel to the line $y = 3x - 1$.

Solution:

Since the slope of the given line equals 3 it suffices to solve $f'(x) = 3$ for x to locate these points(s).

```
sage: var('f, sol') 404
```

```
(f, sol) 405
```

```
sage: f(x)=x^2*e^(3*x+5)+3*x 406
```

```
sage: sol=solve(diff(f(x))==3,x) 407
```

```
sage: sol 408
```

```
[ 409
```

```
x == (-2/3), 410
```

```
x == 0 411
```

```
] 412
```

Hence, there are two solution: $(x_1, f(x_1))$ and $(x_2, f(x_2))$:

```
sage: var('x1, x2') 413
```

```
(x1, x2) 414
```

```
sage: x1=sol[0].rhs() 415
```

```
sage: x2=sol[1].rhs() 416
```

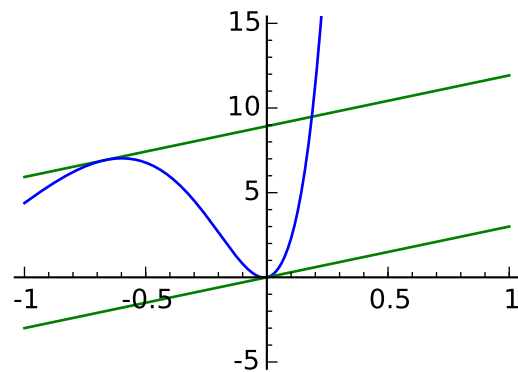
```
sage: f(x1) 417
```

```
4/9*e^3 - 2 418
```

```
sage: f(x2) 419
0 420
```

The plot below confirms that the two corresponding tangent lines (in green) are indeed parallel.

```
sage: y1= f(x1)+(diff(f(x)).substitute(x=x1))*(x-x1) 421
sage: y2= f(x2)+(diff(f(x)).substitute(x=x2))*(x-x2) 422
sage: g1=plot(y1,x,-1,1,color='green',figsize=3,ymin=-5,ymax 423
=15)
sage: g2=plot(f(x),x,-1,1,figsize=3,ymin=-5,ymax=15) 424
sage: g3=plot(y2,x,-1,1,color='green',figsize=3,ymin=-5,ymax 425
=15)
```



Chapter 4

Applications of the Derivative

We have seen how the derivative of a function is itself a function. This idea leads to many possible applications, some of which we will now explore with Sage to demonstrate its ability to manipulate and calculate complicated or tedious expressions.

4.1 Related Rates

Also notice that Sage will display the first derivative of function $S(t)$ as:

$$D[0](S)(t) = \text{diff}(S(t))$$

Example 4.1.1. Let us assume a rubber ball is sitting out in the sun and that the heat causes its surface area the increase at the rate of 3 square centimeters per hour. How fast is the radius increasing when the radius is 2 centimeters?

To solve this problem, we will need the formula for the surface area of a sphere: $S = 4\pi r^2$. Here, the surface area S and the radius r are expressed as functions of t (time).

```
sage: var('t,S,r') 426
```

```
(t, S, r) 427
```

```
sage: r(t)=function('r',t) 428
```

```

sage: S(t)=function('S',t) 429
sage: sa=S(t)==4*pi*(r(t))^2 430
sage: dsa=diff(sa,t) 431
sage: dsa 432
D[0](S)(t) == 8*pi*r(t)*D[0](r)(t) 433

```

Now differentiate this formula and solve for $r'(t)$:

```

sage: sol=solve(dsa, diff(r(t))) 434
sage: sol 435
[ 436
D[0](r)(t) == 1/8*D[0](S)(t)/(pi*r(t)) 437
] 438

```

Since the output above is a nested list (each set of square braces denotes a list) and our solution, $\frac{S'(t)}{8\pi r(t)}$, represent the second element of the first list, we can extract it in order to define $r'(t)$ as follows:

```

sage: Dr(t)=sol[0].rhs() 439
sage: Dr(t) 440
1/8*D[0](S)(t)/(pi*r(t)) 441

```

Note: we will use $Df(x)$ to denote the first derivative of function $f(x)$. As above, $Dr(t) = r'(t)$.

Since we are given that $S'(t) = 3$ and $r(t) = 2$, we substitute these into the formula for $r'(t)$:

```

sage: n(Dr(t).substitute(r(t)==2, diff(S(t),t)==3), digits=3) 442
0.0597 443

```

Therefore, when the radius is 2, it is increasing at the rate of about 0.0597 cm per hour.

4.2 Extrema

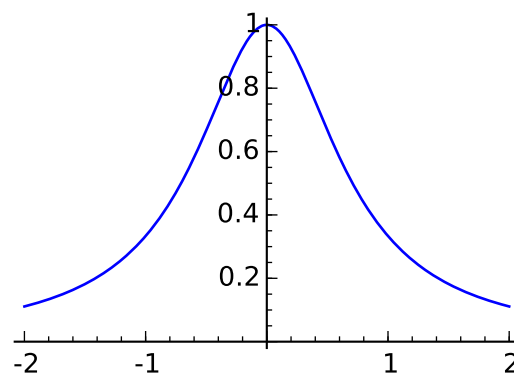
We now consider how to find critical points and inflection points to determine extrema. Recall that critical points of a function are those for which $f'(x) = 0$ or for which $f'(x)$ does not exist. Similarly, inflection points occur where either $f''(x) = 0$ or where $f''(x)$ does not exist. Extrema occur at critical points, but not all critical points are extrema. An inflection point is a point $(c, f(c))$ where concavity changes; this occurs where $f''(c) = 0$ or where $f''(x)$ does not exist, and like critical points, not all points where $f''(x) = 0$ (or where $f''(x)$ does not exist) are inflection points.

Example 4.2.1. Find all local extrema and inflection points of $f(x) = 1/(x^2 + 1)$

Solution:

We first define $f(x)$ in Sage:

```
sage: var('x, f') 444
(x, f) 445
sage: f(x)=1/(2*x^2+1) 446
sage: g=plot(f(x), x, -2, 2, figsize=3, ymin=0, ymax=1) 447
```



To find extrema of f , we locate its critical points, that is, those points where $f'(x) = 0$ or $f'(x)$ is undefined. We can solve the first case using Sage:

```
sage: Df(x)=diff(f(x), x) 448
sage: Df(x) 449
```

```

-4*x/(2*x^2 + 1)^2                                450
sage: solve(Df(x)==0,x)                             451
[                                                    452
x == 0                                              453
]                                                    454

```

Since $f'(x)$ is defined everywhere, it follows that there is exactly one critical point at $x = 0$, and at that point, there is a maximum, as can be seen from the graph above. We could also have used the second derivative test to confirm this:

```

sage: diff(f(x),x,2).substitute(x=0)                455
-4                                                    456

```

Since the second derivative is negative at $x = 0$, the curve is concave down there. This means that we have a local maximum at $x = 0$.

To find the points of inflection, we locate zeros of the second derivative:

```

sage: solve(diff(f(x),x,2)==0,x)                    457
[                                                    458
x == -1/6*sqrt(6),                                  459
x == 1/6*sqrt(6)                                    460
]                                                    461

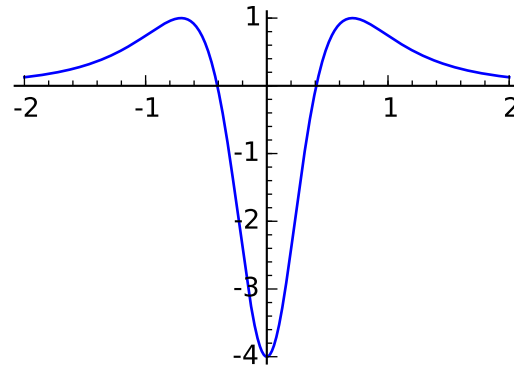
```

To determine if these solutions are indeed inflection points, we need to check if there is a sign change in $f''(x)$ on either side of each.

```

sage: g=plot(diff(f(x),x,2),x,-2,2,figsize=3)       462

```



Notice from the graph above that $f''(x)$ changes from positive to negative at $x = -\frac{\sqrt{6}}{6}$ and from negative to positive at $x = \frac{\sqrt{6}}{6}$. Thus, both point $(-\frac{\sqrt{6}}{6}, f(-\frac{\sqrt{6}}{6}))$ and $(\frac{\sqrt{6}}{6}, f(\frac{\sqrt{6}}{6}))$ are inflection points.

4.3 Optimization

Extreme values of a function occur where the first derivative $f'(x) = 0$ or $f'(x)$ does not exist. This idea allows us to find maximum and minimum, a very important and widely applied in many applications. For example, in business, people want to maximize the profits and minimize the costs. In auto industry, we want to know what shape of the car will minimize the air resistant. There are many similar problems exist in many other fields. We will go over some of these applications in this chapter.

4.3.1 Traffic Flow

Example 4.3.1. Traffic flow along a major highway in Philly between 6 AM and 10 AM can be modeled by the function $f(t) = 20t - 40\sqrt{t} + 50$ (in miles per hour), where $t = 0$ corresponds to 6 AM. Determine when the minimum traffic flow occurs.

Solution:

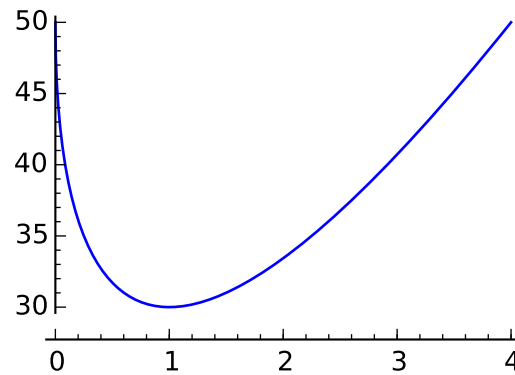
Let us find plot the graph of $f(t)$

```
sage: var('f,t')
```

```
(f, t) 464
```

```
sage: f(t)= 20*t-40*sqrt(t)+50 465
```

```
sage: g=plot(f(t),t,0,4,figsize=3) 466
```



Note from the plot above that the average speed is decreasing between 6 AM to 7 AM and increasing after 7 AM.

At 6 AM the average speed is:

```
sage: f(0) 467
```

```
50 468
```

or 50 mph. At 7 AM the average speed is:

```
sage: f(1) 469
```

```
30 470
```

or 30 mph. To see how the average speed varies throughout the day we make a table of these values at each half hour from 6 AM to 10 AM:

```
sage: step=float(1/2) 471
```

```
sage: initial=float(0) 472
```

```
sage: table([(i*step+initial,n(f(i*step+initial),digits=4)) 473
```

```
for i in [0..8]],align='left')
```

```
0.0    50.00 474
```

0.5	31.72	475
1.0	30.00	476
1.5	31.01	477
2.0	33.43	478
2.5	36.75	479
3.0	40.72	480
3.5	45.17	481
4.0	50.00	482

We can see from the table that the average speed quickly drops from 50 mph to 30 mph in the first hour and then gradually increases back up to 50 mph during the next 3 hours. If we want to verify that the minimum occurs at 7 AM (or $t = 1$), we can use calculus. Since extrema occur where the derivative is 0, we set the derivative equal to zero and solve for t :

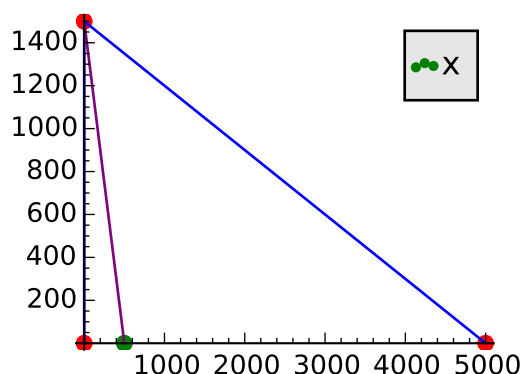
```
sage: solve(diff(f(t),t)==0,t) 483
[ 484
t == 1 485
] 486
```

Therefore the minimum does occur when $t = 1$ (at 7 AM) and from the table we see that the minimum average speed at this time is 30 mph.

4.3.2 Minimum Cost

Example 4.3.2. Imagine there is an island located at $(0, 1500)$ and a mainline electronic connection point at $(5000, 0)$ where the unit is in meter. What would be the cheapest way to connect the island and mainland if the cost to lay cable underwater is 36 and on land is 24? We can lay cable underwater from $(1500, 0)$ to $(x, 0)$ and then lay cable on land from $(x, 0)$ to $(5000, 0)$. The variable x can vary between 0 and 5000. What value of x would minimize the cost for laying this cable and what would that minimum cost be?

Solution:



First, we need to determine the cost. There are two parts: the underwater part and the overland part. The cost of underwater part called c_1 is \$36 times the distance d_1 from $(0, 1500)$ to $(x, 0)$:

```
sage: var('x, c1') 487
```

```
(x, c1) 488
```

```
sage: c1(x)=36*sqrt(1500^2+x^2) 489
```

The overland cost called c_2 is \$24 times the distance d_2 from $(x, 0)$ to $(5000, 0)$:

```
sage: var('x, c2') 490
```

```
(x, c2) 491
```

```
sage: c2(x)=24*(5000-x) 492
```

The total cost is:

```
sage: var('x, cost') 493
```

```
(x, cost) 494
```

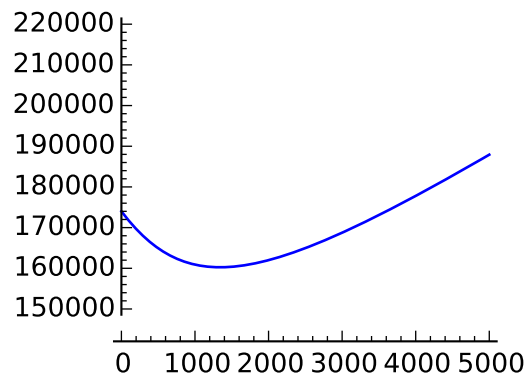
```
sage: cost(x) = c1(x)+c2(x) 495
```

```
sage: cost(x) 496
```

```
-24*x + 36*sqrt(x^2 + 2250000) + 120000 497
```

We need to minimize this cost function. First, we graph it to see if it has a minimum:

```
sage: g=plot(cost(x),x,0,5000,ymin=150000,ymax=220000,figsize=498
      =3)
```



Notice that this cost function has its minimum somewhere between 1000 and 2000. Also, we will note that as x gets close to that minimum the tangent lines of $\text{cost}(x)$ are getting close to horizontal. In other words, the minimum will occur at a point x for which the derivative is zero or horizontal. This is a calculus problem that we can solve.

Also notice that in this particular problem, **solve** command will not evaluate the solution. We have to use **find_root** to numerically approximate the solution:

```
sage: var('c') 499
c 500
sage: c=find_root(diff(cost(x)),0,10000) 501
sage: c 502
1341.6407865 503
sage: n(cost(c)) 504
160249.223594996 505
```

The minimum occurs at $x = 1341.64$ meters and minimum cost is approximately \$160,250

4.3.3 Packaging (Minimum Surface Area)

Example 4.3.3. The cost of packaging in business is related to the surface area of the package. Minimizing the surface area will minimize the cost. Assuming that a Samsung has a refrigerator product that needs to be packaged in a rectangular box having a square base. If the volume of the box is required to be 2 cubic meter, then find the dimensions of the box that will minimize its surface area.

Solution:

Let sides of the square base is x and the height of the box is y , then the volume of the box is given by x^2y and must equal 2 cubic meters.

```
sage: var('x,y,S') 506
```

```
(x, y, S) 507
```

```
sage: constraint=x^2*y==2 508
```

The surface area of the box is $S = 4xy + 2x^2$ and is the quantity that must be minimized, where the area of top and bottom sides are x^2 and the 4 sides each have area xy . Using our volume constraint, $x^2y = 2$, we can solve for y in terms of x :

$$y = \frac{2}{x^2}$$

```
sage: sol=solve(constraint,y) 509
```

```
sage: sol 510
```

```
[ 511
```

```
y == 2/x^2 512
```

```
] 513
```

The surface area function can then be expressed as a function of x only:

$$S(x) = 4xy + 2x^2 = 4x(2/x^2) + 2x^2 = 8/x + 2x^2$$

```
sage: S(x)=(4*x*y+2*x^2).substitute(y==sol[0].rhs())
```

514

```
sage: S(x)
```

515

```
2*x^2 + 8/x
```

516

Again, we use the idea that extrema occur at points where the derivative is zero, we have:

```
sage: solve(diff(S(x),x)==0,x)
```

517

```
[
```

518

```
x == 1/2*I*sqrt(3)*2^(1/3) - 1/2*2^(1/3),
```

519

```
x == -1/2*I*sqrt(3)*2^(1/3) - 1/2*2^(1/3),
```

520

```
x == 2^(1/3)
```

521

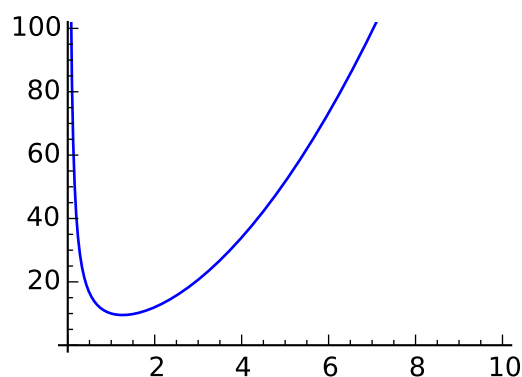
```
]
```

522

This equation has 1 real and 2 imaginary solutions. We need only the real solution of $x = 2^{1/3}$. We compare with the plot to see the actual minimum:

```
sage: g=plot(S(x),x,0,10,ymin=0,ymax=100,figsize=3)
```

523



Alternatively, we could have used the second derivative test to show that a minimum occurs at $x = 2^{1/3}$:

```
sage: (diff(S(x),x,2)).substitute(x==2^(1/3))
```

524

```
12
```

525

Since $f''(2^{1/3}) > 0$, we know that the graph is concave up at $x = 2^{1/3}$ and hence must have a

minimum there. Since $y = 2^{1/3}$ when $x = 2^{1/3}$, we conclude that the box with minimum surface area is a 2 cube meters with sides of $2^{1/3}$ meters.

4.3.4 Maximize Revenue

Example 4.3.4. Suppose a travel agency charges 500 per person for a charter flight if exactly 80 people sign up. However, if more than 80 people sign up, then the fare is reduced by 2 per person for each additional person over the initial 80. The travel agency wants to know how many people they should book to maximize revenue. Also, determine what that maximum revenue is and what the corresponding fare is for each person.

Solution:

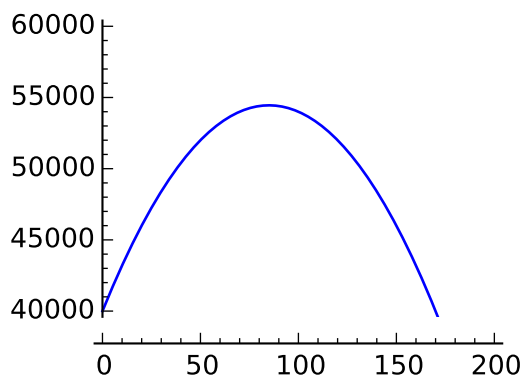
Let x denotes the number of passenger above 80 and the revenue is the product of the number of people multiplied by the cost (fare) per person. If $R(x)$ is defined as the revenue function, then $R(x) = (80 + x)(500 - 2x)$. We want to determine the maximum value of $R(x)$ for $x \geq 0$. Let consider the graph:

```
sage: var('x,R') 526
```

```
(x, R) 527
```

```
sage: R(x)=(80+x)*(500-2*x) 528
```

```
sage: g=plot(R(x),x,0,200,ymin=40000,ymax=60000,figsize=3) 529
```



From the plot above, we see that a maximum occurs at about 80 to 90. To confirm this, we first

solve for the critical points:

```
sage: solve(diff(R(x),x)==0,x) 530
[ 531
x == 85 532
] 533
```

Therefore the maximum does indeed occur at $x = 85$, and the maximum revenue is:

```
sage: R(85) 534
54450 535
```

or \$54450. Since $80 + x$ represents the number of customers, this occurs when 165 customers sign up for the flights. In this case, the cost per person is:

```
sage: (500 - 2*x).substitute(x==85) 536
330 537
```

or \$330 per person.

4.4 Newton's Method

4.4.1 Programing Newton's Method

Newton's Method is a technique for calculating zeros of a function based on the direction of its tangent lines (hence, it requires first derivative). It is a recursive routine. tedious to do by hand and easily to make mistake. However, it is simple to handle with Sage. We need initial guess value to start with or in other word, we need to guess where to solution's location is. This is because an initial approximation x_0 for that zero, say at $x = r$, is needed to start the recursion. For example, we can specify x_0 by examining the graph of the function to see where the zeros are approximately. Then the next approximation x_1 can be found by the recursive formula $x_1 = x_0 - f(x_0)/f'(x_0)$.

This process can be iterated using the general formula:

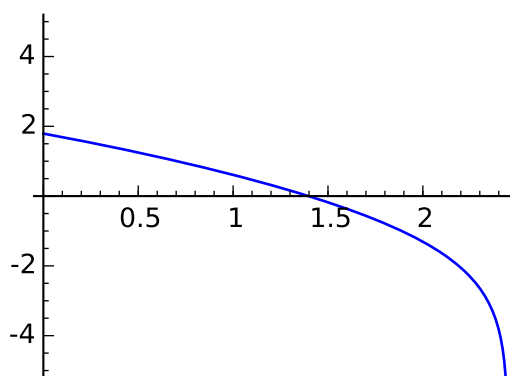
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Under suitable conditions, the sequence of approximation $\{x_0, x_1, x_2, \dots\}$ (called Newton sequence) will converge to r . However the Newton Method does not guarantee the convergent, if the initial guess is not good (or not close enough to the zero) then it will diverges, meaning we will not able to find the solution.

Example 4.4.1. Approximate the zeros of the function $f(x) = \ln(6 - x^2) - x$.

Solution:

```
sage: var('x,f') 538
(x, f) 539
sage: f(x)=log(6-x^2)-x 540
sage: g=plot(f(x),x,0,4,ymin=-5,ymax=5,figsize=3) 541
```



Clearly, there is one zero between 1 and 1.5 based on the graph above. To approximate this zero, we define a function **newtn** to perform the recursion:

```
sage: var('x,newtn') 542
(x, newtn) 543
```

```
sage: newtn(x)=x-f(x)/(diff(f(x)))
```

544

To generate the corresponding Newton sequence, we compute 8 iterates of this function starting with an initial guess of $x = 1.5$.

```
sage: xzero=float(15/10)
```

```
sage: for i in range(8):
```

```
....:     xzero=newtn(xzero)
```

```
....:     print xzero
```

```
1.4009754666568441
```

```
1.3977834736657635
```

```
1.3977805354266575
```

```
1.3977805354241768
```

```
1.397780535424177
```

```
1.3977805354241768
```

```
1.397780535424177
```

```
1.3977805354241768
```

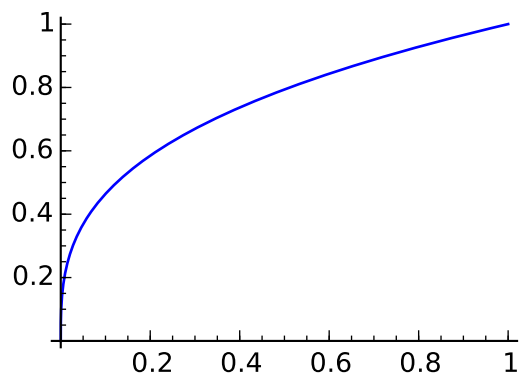
Hence, if we stop at 6 decimal spaces then the zero of $f(x) = \ln(6 - x^2) - x$ is 1.397780.

4.4.2 Divergence

As mention earlier, Newton's Method does not alway work. For instance, the function $y = x^{1/3}$ clearly has a root at $x = 0$:

```
sage: g=plot(x^(1/3),x,0,1,figsize=3,ymin=0,ymax=1)
```

545



Yet, Newton's Method fails for any guess $x \neq 0$:

```
sage: f(x) = x^(1/3)
```

```
sage: newtn(x)=x-f(x)/(diff(f(x)))
```

```
sage: xzero=float(5/10)
```

```
sage: for i in range(8):
```

```
....:     xzero=newtn(xzero)
```

```
....:     print xzero
```

```
-1.0
```

```
2.0
```

```
-4.0
```

```
8.0
```

```
-16.0
```

```
32.0
```

```
-64.0
```

```
128.0
```

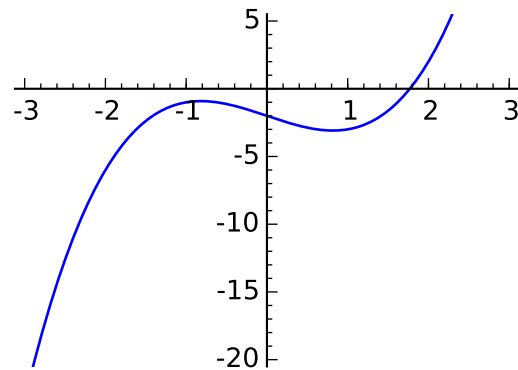
4.4.3 Slow Convergence

Even when Newton's Method works, sometimes the Newton sequence converges very slowly to the zero. Consider the following function:

```

sage: var('x,f')
(x, f)
sage: f(x)=x^3-2*x-2
sage: g=plot(f(x),x,-3,3,ymin=-20,ymax=5,figsize=3)

```



Clearly, there is a root between 1.5 and 2. If we use the **newtn** function with our guess at $x = 1$, we get quick convergence to root:

```

sage: f(x)=x^3-2*x-2
sage: newtn(x)=x-f(x)/(diff(f(x)))
sage: xzero=float(1)
sage: for i in range(8):
...:     xzero=newtn(xzero)
...:     print xzero
4.0
2.8260869565217392
2.1467190137392356
1.8423262771400926
1.772847636439238
1.7693013974364495
1.7692923542973595

```

1.7692923542386314

But if we choose our initial guess near 0.7, the convergence is much slower. (It took 20 iterations to have the accuracy as the 8th iteration above).